



Caringbah High School

Year 12 2025

Mathematics Extension 1

HSC Course

Assessment Task 4 Trial Examination

General Instructions

- Reading time – 10 minutes
- Working time – 2 hours
- Write using black pen
- NESA-approved calculators may be used
- A reference sheet is provided
- In Questions 11-14, show relevant mathematical reasoning and/or calculations
- Marks may not be awarded for partial or incomplete answers

Total marks – 70

Section I 10 marks

Attempt Questions 1-10
Mark your answers on the answer sheet provided. You may detach the sheet and write your name on it.

Section II 60 marks

Attempt Questions 11-14
Write your answers in the answer booklets provided. Ensure your name or student number is clearly visible.

Student Number: _____

Class: _____

Marker's Use Only						
Section I	Section II				Total	
Q 1-10	Q11	Q12	Q13	Q14		
/10	/15	/15	/15	/15	/70	%

Section I

10 marks

Attempt Questions 1-10

Allow about 15 minutes for this section

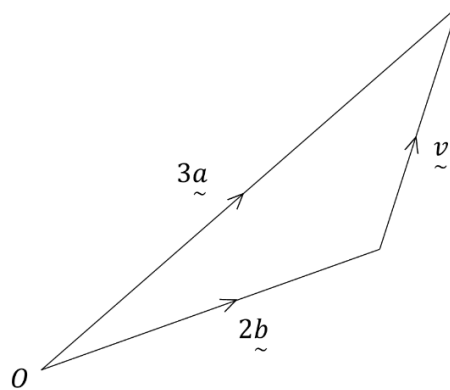
Use the multiple-choice answer sheet for Questions 1–10

1. What is the solution of the inequality $x(x - 4) < 5$?
 - (A) $(0, 4)$
 - (B) $(-1, 5)$
 - (C) $(-\infty, -1) \cup (5, \infty)$
 - (D) $(-\infty, 0) \cup (4, \infty)$

2. What are the domain and range of the function $f(x) = \cos^{-1}\left(\frac{x}{4}\right)$?
 - (A) Domain: $-1 \leq x \leq 1$, Range: $0 \leq y \leq \pi$
 - (B) Domain: $-4 \leq x \leq 4$, Range: $0 \leq y \leq 4\pi$
 - (C) Domain: $-4 \leq x \leq 4$, Range: $0 \leq y \leq \pi$
 - (D) Domain: $-1 \leq x \leq 1$, Range: $0 \leq y < 4\pi$

3. $(x - 1)$ is a factor of the polynomial $P(x) = x^3 + kx^2 + 3x - 2$, for some constant k .
What is the remainder when $P'(x)$ is divided by $(x - 1)$?
 - (A) -2
 - (B) -1
 - (C) 1
 - (D) 2

4. It is given that $\underline{a} = \overrightarrow{OA} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ and $\underline{b} = \overrightarrow{OB} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$, and $\underline{v}$ starts at the tip of $2\underline{b}$ and finishes at the tip of $3\underline{a}$ as shown.



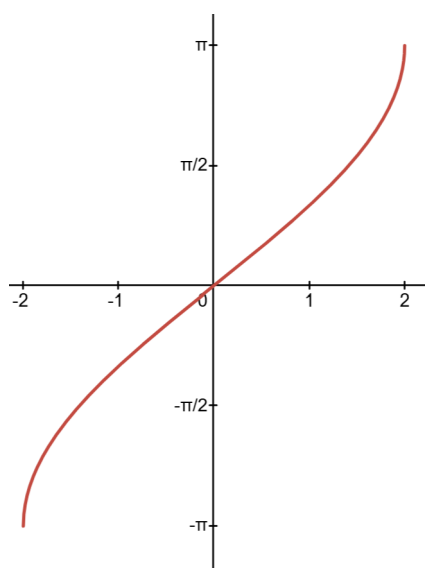
Which vector below is equal to $\underline{v}$?

- (A) $\begin{pmatrix} -2 \\ 3 \end{pmatrix}$
- (B) $\begin{pmatrix} 2 \\ -3 \end{pmatrix}$
- (C) $\begin{pmatrix} 4 \\ 4 \end{pmatrix}$
- (D) $\begin{pmatrix} 10 \\ 9 \end{pmatrix}$
5. Given that $\cos x - 2\sin x + 2 = 0$. What are the values of t , if $t = \tan\left(\frac{x}{2}\right)$?
- (A) $t = 1$ or $t = 3$
- (B) $t = 1$ or $t = -2$
- (C) $t = 1$ or $t = 2$
- (D) $t = 1$ or $t = -3$

6. Fifteen cards are numbered 1 to 15 and placed on a table so that their numbers are hidden. The cards are flipped over one at a time. What is the minimum number of cards that must be flipped to obtain at least one pair with a difference of 7?

- (A) 7
(B) 8
(C) 9
(D) 10

7. The diagram shows the graph of a function. Which function does the graph represent?



- (A) $y = 2\sin^{-1}(2x)$
(B) $y = -2\cos^{-1}(2x)$
(C) $y = 2\sin^{-1}\left(\frac{x}{2}\right)$
(D) $y = 2\cos^{-1}\left(\frac{x}{2}\right)$

8. What is the primitive of $\frac{3 + 2x}{4 + x^2}$?

(A) $\frac{3}{2}\tan^{-1}(x) + \ln(4 + x^2) + c$

(B) $3\tan^{-1}\left(\frac{x}{2}\right) + \ln(4 + x^2) + c$

(C) $\frac{3}{2}\tan^{-1}(x) + \ln(x^2) + c$

(D) $\frac{3}{2}\tan^{-1}\left(\frac{x}{2}\right) + \ln(4 + x^2) + c$

9. The coefficient of x^2 in the expansion of $\left(5x + \frac{2}{x}\right)^8$ is:

(A) 112 000

(B) 1 400 000

(C) 700 000

(D) 224 000

Section I continues next page

10. In a game of basketball, Sam has a scoring accuracy of 70%. What is the ratio of Sam scoring at least one of three attempts compared to scoring exactly twice out of three attempts?

- (A) 3: 2
- (B) 973: 189
- (C) 49: 3
- (D) 139: 63

End of Section I

Section II

60 marks

Attempt Questions 11–14

Allow about 1 hour 45 minutes for this section

Answer each question in a SEPARATE writing booklet.

Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet.

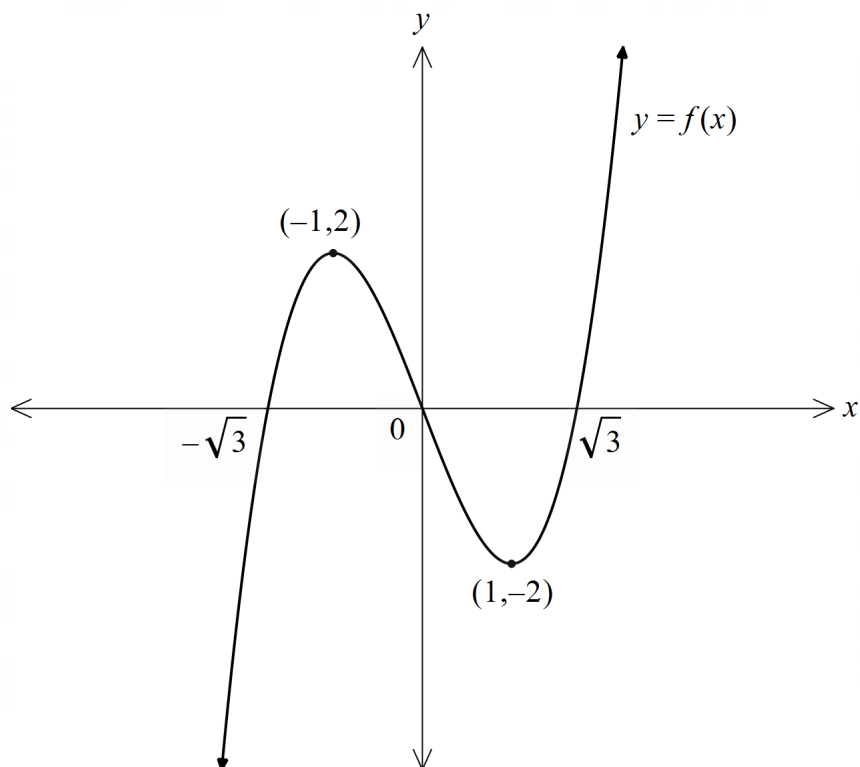
- (a) In how many ways can the letters of the word SUCCESS be arranged in a line if:
- (i) there are no restrictions? 1
 - (ii) the two Cs are NOT adjacent? 2
- (b) Use division of polynomials to express $P(x)$ in the form 2
 $P(x) = A(x)Q(x) + R(x)$, where $P(x) = 3x^3 + 14x + 11$ and $A(x) = x^2 - 3x + 2$.
- (c) Use mathematical induction to prove that $8^n - 1$ is divisible by 7, for all 3
integers $n \geq 1$.
- (d) $\underline{a}$ and $\underline{b}$ are two non-zero vectors. Prove that the vectors $\underline{u}$ and $\underline{v}$ below are 2
perpendicular to each other.
- $$\underline{u} = |\underline{b}|\underline{a} + |\underline{a}|\underline{b}$$
- $$\underline{v} = |\underline{b}|\underline{a} - |\underline{a}|\underline{b}$$
- (e) Using the substitution $u = x - 1$, or otherwise, find 2
- $$\int (x + 3)(x - 1)^5 dx$$

Question 11 continues next page

- (f) The diagram below shows the graph of $y = f(x)$.

3

Sketch $y = \frac{1}{f(x)}$ **over the graph in template provided**, showing all turning points, relevant coordinates, and asymptotes.



End of Question 11

Question 12 (15 marks) Use a SEPARATE writing booklet.

(a) Given $\underline{u} = 4\underline{i} + 3\underline{j}$ and $\underline{v} = 3\underline{i} + 5\underline{j}$. Find

- (i) $\underline{u} \cdot \underline{v}$ 1
- (ii) the angle between $\underline{u}$ and $\underline{v}$ to the nearest degree. 2

(b) (i) Given $\cos x + \sqrt{3} \sin x = A \cos(x - \alpha)$, find A and α , where $0 \leq \alpha \leq 2\pi$ and $A > 0$. 2

(ii) Hence, or otherwise, solve $\cos x + \sqrt{3} \sin x = 2$, $0 \leq x \leq 2\pi$. 2

(c) A spherical bubble is moving up through a liquid. As it rises, the bubble gets bigger, and its radius increases at a rate of 0.3 mm/s. 2

At what rate is the volume of bubble increasing when its radius reaches 0.7 mm, give your answer correct to one decimal place.

(d) The region bounded by the curve $y = \sqrt{x}$, the y -axis and the line $y = 2$ is rotated about the y -axis. Calculate the exact volume of the solid of revolution. 2

(e) A factory produces computer chips. Each chip has a 2% chance of being faulty and is produced independently. The factory's quality control department tests a random sample of 100 chips.

(i) Calculate the mean and variance of the random variable $X \sim \text{Bin}(100, 0.02)$. 2

(ii) Using a binomial distribution, calculate the probability that a maximum of three chips is faulty. Give your answer correct to four decimal places. 2

End of Question 12

Question 13 (15 marks) Use a SEPARATE writing booklet.

(a) A four-person team is to be chosen at random from nine women and seven men.

(i) In how many ways can the team be chosen. 1

(ii) What is the probability that the team will consist of 4 men. 1

(b) Evaluate 3

$$\int_0^{\frac{\pi}{2}} \sin^2(3x) dx$$

(c) The unit vectors $\underline{p}$ and $\underline{q}$ make an angle of $\frac{\pi}{3}$ with each other. 3

Find the magnitude of $\underline{p} - \frac{\underline{q}}{2}$.

(d) Use Mathematical Induction to prove that 3

$$\cos x + \cos(3x) + \cos(5x) + \dots + \cos((2n-1)x) = \frac{\sin(2nx)}{2\sin x},$$

for all positive integers $n \geq 1$

Question 13 continues next page

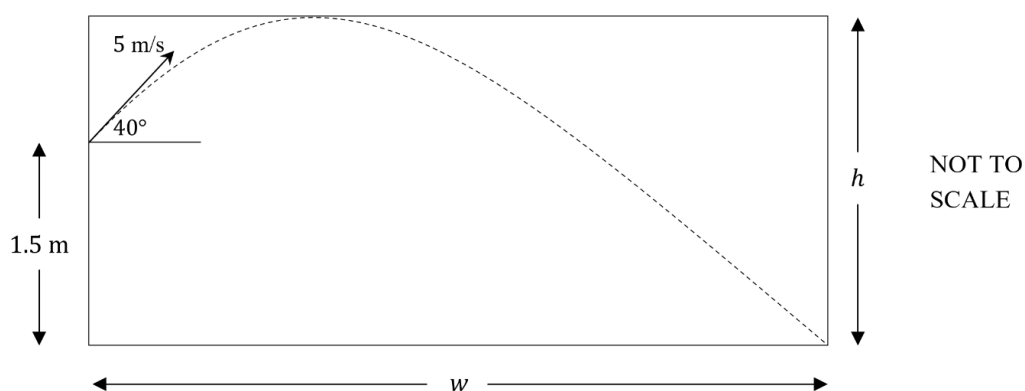
- (e) A student is standing in the doorway to a classroom, which has a height of h metres and a width of w metres. He throws a ball from a height of 1.5 metres, at a speed of 5 m/s with an angle of projection of 40° . The ball just touches the ceiling before landing at the junction of the floor and the wall, as shown in figure below.

The horizontal and vertical displacement t seconds after projection are:

$$x = 5t\cos(40^\circ)$$

$$y = 5t\sin(40^\circ) - 5t^2 + 1.5$$

(DO NOT PROVE THESE)



Find the width and height of the room in metres, correct to 1 decimal place.

4

End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet.

- (a) Given that k is a non-zero constant and n is a positive integer, then **3**

$$(1 + kx)^n = 1 + 40kx + 120k^2 x^2 + \dots$$

Find the values of k and n .

- (b) It is known that $\cos(4\theta) = 8\cos^4\theta - 8\cos^2\theta + 1$ (DO NOT PROVE THIS)

- (i) Find the exact values of the solutions to the equation **2**

$$8x^4 - 8x^2 + 1 = 0, \text{ for } 0 \leq \theta \leq \pi$$

- (ii) Hence, or otherwise, find the exact value of $\cos\frac{\pi}{8}\cos\frac{3\pi}{8}$. **2**

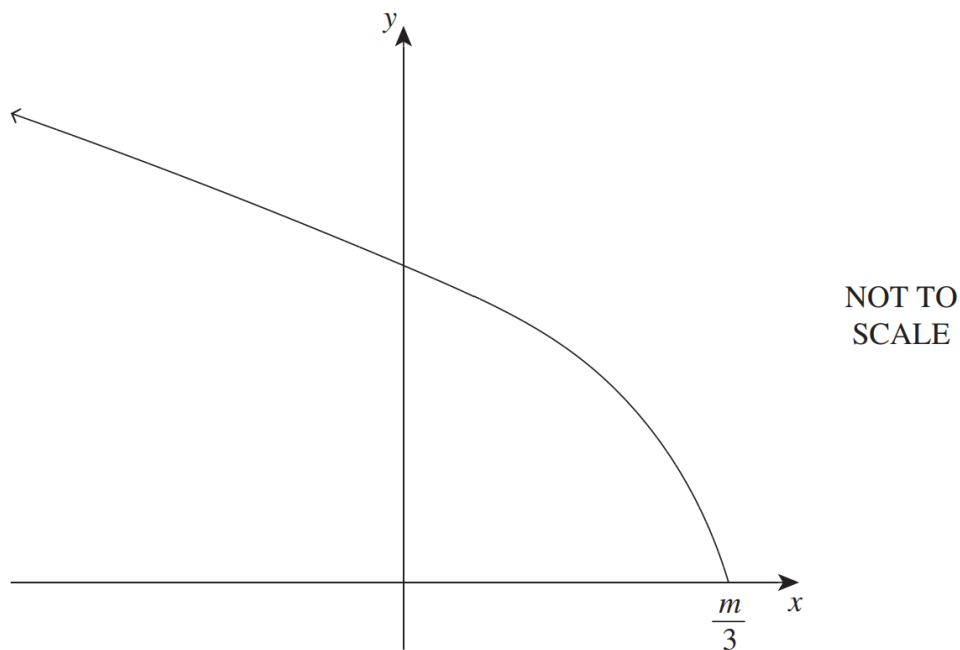
- (c) Use the substitution of $x = \tan^2(\theta)$, $0 \leq \theta < \frac{\pi}{2}$, to evaluate: **4**

$$\int_0^1 \frac{1-x}{\sqrt{x}(1+x)^2} dx$$

Question 14 continues next page

- (d) Let $f(x) = \sqrt{m-3x}$ for $x \leq \frac{m}{3}$. The graph of $y = f(x)$ is shown below.

4



The area enclosed by the graph $y = f(x)$, the x -axis and the y -axis is rotated about the y -axis.

Find the value of m such that the volume of the solid formed is $\frac{5000\pi}{27}$ cubic units.

End of Examination

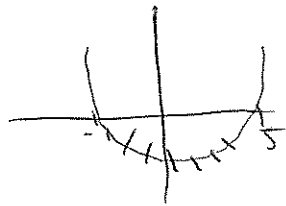
Multiple Choice

1) $x(x-4) < 5$

$$x^2 - 4x - 5 < 0$$

$$(x-5)(x+1) < 0$$

$$-1 < x < 5 - \textcircled{B}$$



2) $-1 \leq \frac{2x}{4} \leq 1$

$$-4 \leq x \leq 4$$

$$\text{Range} = 0 \leq y \leq \pi - \textcircled{C}$$

3) $P(1) = 0$

$$1 + k + 3 - 2 = 0$$

$$k = -2$$

$$P'(x) = 3x^2 + 2kx + 3$$

$$P'(x) = 3x^2 - 4x + 3$$

$$P'(1) = 3 - 4 + 3 = 2$$

$$- \textcircled{D}$$

4) $\underline{v} = -2\underline{b} + 3\underline{a}$

$$= -2\left(\frac{2}{3}\right) + 3\left(\frac{2}{1}\right)$$

$$= \begin{bmatrix} -4 & +6 \\ -6 & +3 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 3 \\ -3 & \end{bmatrix} - \textcircled{B}$$

5) $\frac{1-t^2}{1+t^2} - 2\left(\frac{2t}{1+t^2}\right) + 2 = 0$

$$1-t^2 - 4t + 2(1+t^2) = 0$$

$$t^2 - 4t + 3 = 0$$

$$(t-3)(t-1) = 0$$

$$t = 3 \quad \text{or} \quad t = 1 - \textcircled{A}$$

6) (1,8), (2,4), (3,10), (4,17), (5,24), (6,31), (7,38), (8,45)

By PGF $\underline{a} \Rightarrow \textcircled{C}$

7) $\sin^{-1}x$ dilated by a factor of

- 2 Horizontally

- 2 Vertically

$$\Rightarrow y = \sin^{-1}\left(\frac{x}{2}\right)$$

$$y = 2\sin^{-1}\left(\frac{x}{2}\right) \Rightarrow \textcircled{C}$$

$$8) \int \frac{3+2x}{4+x^2}$$

$$= \int \frac{3}{4+x^2} + \int \frac{2x}{4+x^2}$$

$$= \frac{3}{2} \tan^{-1}\left(\frac{x}{2}\right) + \ln(4+x^2) + C$$

$$\Rightarrow \textcircled{D}$$

$$9) {}^8C_r \left(5x\right)^r \left(\frac{2}{x}\right)^{8-r}$$

$$= {}^8C_r 5^r x^r \times 2^{8-r} x^{r-8}$$

$$= {}^8C_r \times 5^r \times 2^{8-r} x^{2r-8}$$

$$\Rightarrow {}^8C_5 \times 5^5 \times 2^{8-5}$$

$$= 1400000$$

$$\Rightarrow \textcircled{B}$$

$$2r-8=2$$

$$2r=10$$

$$r=5$$

(10) $X \sim (3, 0.1)$

$$\begin{aligned} \bullet P(X \geq 1) &= 1 - P(X=0) \\ &= 1 - {}^3C_0 (0.7)^0 (0.3)^3 \\ &= 1 - 0.027 \\ &= 0.973 \end{aligned}$$

$$\begin{aligned} \bullet P(X=2) &= {}^3C_2 (0.7)^2 (0.3) \\ &= 0.441 \end{aligned}$$

$$\bullet \frac{0.973}{0.441} = \frac{139}{63} \Rightarrow \textcircled{D}$$

Question 11

a) i) $\frac{7!}{3! \times 2!} = 420$ — ①

ii) $420 - (C_s \text{ are adjacent})$
 $= 420 - \left(\frac{6!}{3!}\right)$ — ①
 $= 360$ — ①

b)

$$\begin{array}{r} \overline{3x+9} \quad \text{--- ①} \\ x^2-3x+2 \left\{ \begin{array}{l} 3x^3+0x^2+14x+11 \\ -(3x^3-9x^2+6x) \\ \hline 9x^2+8x+11 \\ -(9x^2-27x+18) \\ \hline 35x-7 \end{array} \right. \quad \text{--- ①} \end{array}$$

$$P(x) = (x^2-3x+2)(3x+3) + 17x+5$$

c) $\underline{n=1}$: $8^1-1=7$ is divisible by 7 — ①
 $\Rightarrow$ true for $n=1$

* Suppose true for $n=k$, $k \geq 1$, that is

$$8^k-1 = 7m, \text{ for some integer } m \quad \text{--- ①}$$

To prove true for $n=k+1$, that is

$$8^{k+1}-1 = 7p, \text{ for some integer } p$$

$$\begin{aligned} 8^{k+1}-1 &= 8^k \cdot 8 - 1 \\ &= (7m+1) \cdot 8 - 1 \\ &= 56m + 8 - 1 \\ &= 56m + 7 = 7(8m+1) = 7p \end{aligned} \quad \text{--- ①}$$

$\rightarrow$

so by mathematical induction $0 - 1 - 1$ is
divisible by 7, for all integers $n \geq 1$

$$\begin{aligned} d) \quad u \cdot v &= (|b|^2 a + |a|^2 b)(|b|^2 a - |a|^2 b) \quad \text{--- (1)} \\ &= |b|^2 a \cdot a - |a|^2 b \cdot b + |a|^2 b \cdot a - |b|^2 a \cdot b \\ &= |b|^2 |a|^2 - |a|^2 |b|^2 \\ &= 0 \quad \text{--- (1)} \end{aligned}$$

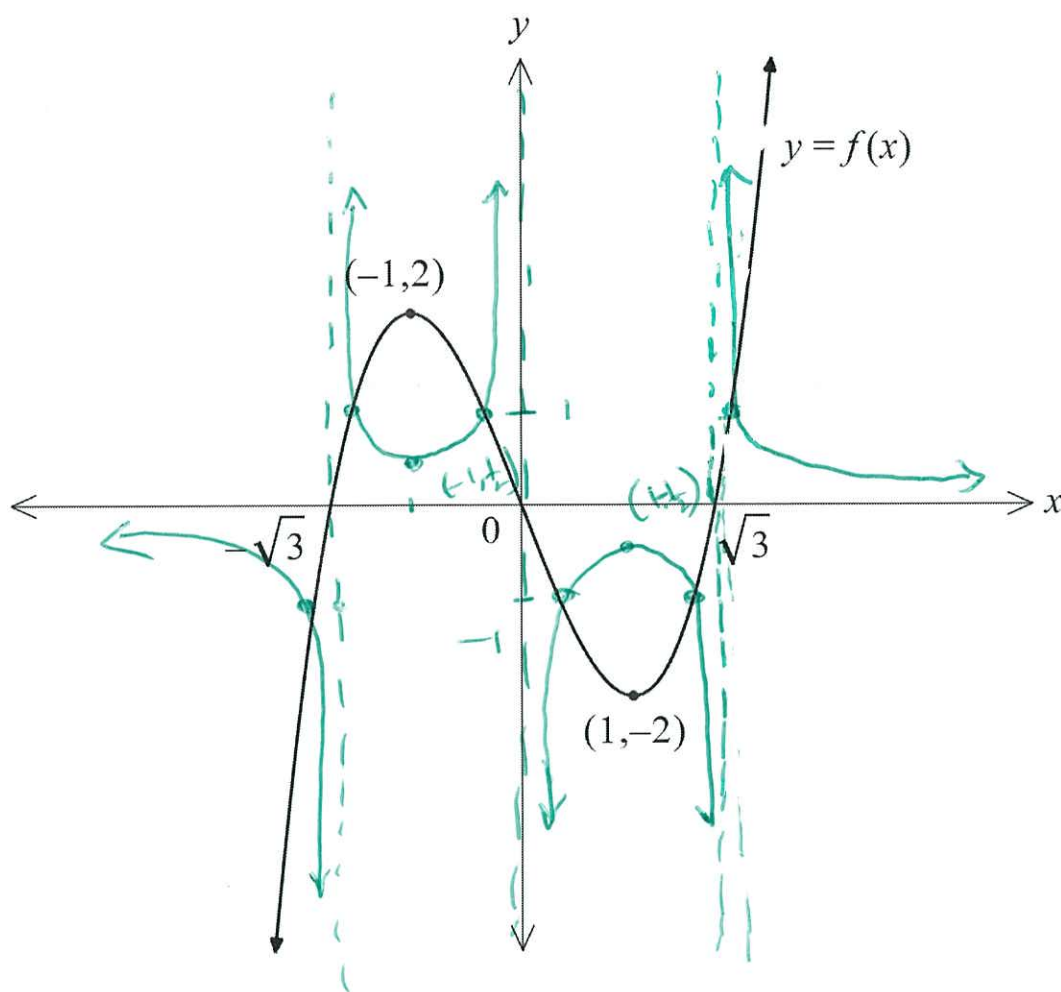
$\Rightarrow u$ and v are perpendicular to each other

$$\begin{aligned} e) \quad u &= x-1 \\ du &= dx \end{aligned}$$

$$\begin{aligned} &\int (u+3)(u-1)^5 du \\ &= \int (u+1+3)(u)^5 du \quad \text{--- (1)} \\ &= \int (u+4)u^5 du \\ &= \int u^6 + 4u^5 du \\ &= \frac{u^7}{7} + \frac{4u^6}{6} + C \quad \text{--- (2)} \\ &= \frac{(x-1)^7}{7} + \frac{2}{3}(x-1)^6 + C \quad \text{--- (1)} \end{aligned}$$

Student Number/Name: _____

Question 11 (f) Template



Asymptote - ①
min/max coord - ①
intersection with original at - ①
1, -1

Question 12:

$$\begin{aligned} \text{a) i) } \underline{u} \cdot \underline{v} &= 4 \times 3 + 3 \times 5 \\ &= 12 + 15 \\ &= 27 \quad - \textcircled{1} \end{aligned}$$

$$\text{ii) } \left. \begin{aligned} |\underline{u}| &= \sqrt{16+9} = 5 \\ |\underline{v}| &= \sqrt{9+25} = \sqrt{34} \end{aligned} \right\} - \textcircled{1}$$

$$\cos \theta = \frac{\underline{u} \cdot \underline{v}}{|\underline{u}| |\underline{v}|}$$

$$\cos \theta = \frac{27}{5\sqrt{34}}$$

$$\boxed{\theta \approx 22^\circ} - \textcircled{1}$$

$$\text{b) i) } \cos x + \sqrt{3} \sin x = A (\cos x \cos \alpha + \sin x \sin \alpha)$$

$$\begin{cases} A \cos \alpha = 1 \\ A \sin \alpha = \sqrt{3} \end{cases}$$

$$A = \sqrt{1+3}$$

$$A = \sqrt{4}$$

$$\boxed{A = 2} - \textcircled{1}$$

$$\sin \alpha = \frac{\sqrt{3}}{2} > 0$$

$$\cos \alpha = \frac{1}{2} > 0$$

$$\Rightarrow \alpha \text{ in Q1}$$

$$\Rightarrow \boxed{\alpha = \frac{\pi}{3}} - \textcircled{1}$$

$$\text{ii) } 2 \cos\left(x - \frac{\pi}{3}\right) = 2 \quad - \textcircled{1}$$

$$\cos\left(x - \frac{\pi}{3}\right) = 1$$

$$u = x - \frac{\pi}{3}$$

$$\cos u = 1$$

$$u = 0, 2\pi$$

$$0 \leq x \leq 2\pi$$

$$\frac{\pi}{3} \leq u \leq \frac{5\pi}{3}$$

$$\Rightarrow x - \frac{\pi}{3} = 0$$

$$\boxed{x = \frac{\pi}{3}} - \textcircled{1}$$

$$c) \quad \frac{dr}{dt} = 0.3 \text{ mm/s}$$

$$\frac{dv}{dt} = ??$$

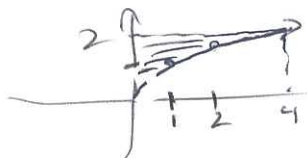
$$\begin{aligned} \frac{dv}{dt} &= \frac{dv}{dr} \times \frac{dr}{dt} \\ &= 4\pi r^2 \times 0.3 \end{aligned}$$

$$\begin{aligned} V &= \frac{4}{3}\pi r^3 \\ \frac{dv}{dr} &= \frac{4}{3} \times 3\pi r^2 \\ \frac{dv}{dr} &= 4\pi r^2 \end{aligned} - \textcircled{1}$$

$$r = 0.7$$

$$\begin{aligned} \Rightarrow \frac{dv}{dt} &= 4\pi \times (0.7)^2 \times 0.3 \\ &= 1.8 \text{ mm}^3/\text{s} - \textcircled{1} \end{aligned}$$

$$d) \begin{cases} y = \sqrt{x} \\ x = y^2 \end{cases}$$



$$\begin{aligned} V &= \pi \int_{-2}^2 [f(y)]^2 dy \\ &= \pi \int_{-2}^2 y^4 dy \\ &= \frac{\pi}{5} [2^5 - 0] \\ &= \frac{32\pi}{5} - \textcircled{1} \end{aligned}$$

c) i) $X \sim \text{Bin}(100, 0.02)$

$$\begin{array}{l|l} \mu = np & \text{Var}(X) = npq \\ = 100(0.02) & = 100(0.02)(0.98) \\ = 2 - \textcircled{1} & = 1.96 - \textcircled{1} \end{array}$$

ii) $P(X \leq 3)$

$$\begin{aligned} \textcircled{1} - \left\{ \begin{aligned} &= P(X=0) + P(X=1) + P(X=2) + P(X=3) \\ &= {}^{100}C_0 (0.02)^0 (0.98)^{100} + {}^{100}C_1 (0.02)^1 (0.98)^{99} \\ &\quad + {}^{100}C_2 (0.02)^2 (0.98)^{98} + {}^{100}C_3 (0.02)^3 (0.98)^{97} \end{aligned} \right. \\ &= 0.8590 - \textcircled{1} \end{aligned}$$

Question 13

$$a) i) {}^{16}C_4 = 1820 - \textcircled{1}$$

$$ii) {}^7C_4 = 35 - \textcircled{1}$$

$$\begin{aligned} b) \quad & \int_0^{\frac{\pi}{2}} \sin^2(3x) dx \\ &= \int_0^{\frac{\pi}{2}} \frac{1 - \cos 2(3x)}{2} \\ &= \int_0^{\frac{\pi}{2}} \frac{1 - \cos 6x}{2} \quad \text{--- } \textcircled{1} \\ &= \left[\frac{1}{2}x - \frac{\sin 6x}{12} \right]_0^{\frac{\pi}{2}} \quad \text{--- } \textcircled{1} \\ &= \left(\frac{\pi}{4} - \frac{\sin 3\pi}{12} \right) - (0 - 0) \\ &= \frac{\pi}{4} \quad \text{--- } \textcircled{1} \end{aligned}$$

$$c) \quad |p| = |q| = 1$$

$$p \cdot q = |p| |q| \cos \frac{\pi}{3}$$

$$p \cdot q = \frac{1}{2} \quad \text{--- } \textcircled{1}$$

$$\left|P - \frac{q}{2}\right|^2 = \left(P - \frac{q}{2}\right)\left(P - \frac{q}{2}\right)$$

$$= |P|^2 - \frac{1}{2}P \cdot q - \frac{1}{2}q \cdot P + \frac{|q|^2}{4} \quad \text{--- (1)}$$

$$= 1 - P \cdot q + \frac{1}{4}$$

$$= 1 - \frac{1}{2} + \frac{1}{4} = \frac{3}{4}$$

$$\Rightarrow \left|P - \frac{q}{2}\right| = \frac{\sqrt{3}}{2} \quad \text{--- (1)}$$

(d) $n=1$ $RHS = \frac{\sin 2x}{2 \sin x} = \frac{2 \sin x \cos x}{2 \sin x} = \cos x = LHS$ } (1)
Hence true for $n=1$

* Suppose true for $n=k$, $k \geq 1$, so

$$\cos x + \cos 3x + \cos 5x + \dots + \cos((2k-1)x) = \frac{\sin 2kx}{2 \sin x}$$

* To prove true for $n=k+1$

$$\cos x + \cos 3x + \cos 5x + \dots + \cos((2k-1)x) + \cos((2(k+1)-1)x) = \frac{\sin 2(k+1)x}{2 \sin x}$$

* $LHS = \underbrace{\cos x + \cos 3x + \dots + \cos((2k-1)x)}_{\frac{\sin 2kx}{2 \sin x}} + \cos((2k+1)x)$

$$= \frac{\sin 2kx}{2 \sin x} + \cos((2k+1)x)$$

$$= \frac{\sin 2kx + 2 \sin x \cos((2k+1)x)}{2 \sin x} \quad \text{--- (1)}$$

$$= \frac{\sin 2kx + \sin((2k+1)x + x) - \sin((2k+1)x - x)}{2 \sin x}$$

$$= \frac{\cancel{\sin 2kx} + \sin(2kx + 2x) - \cancel{\sin(2kx)}}{2 \sin x}$$

$$= \frac{\sin(2(n+1)x)}{2\sin x} \quad \text{--- (1)}$$

$$= R.H.S$$

$\Rightarrow$ By mathematical induction, Statement true for all $n \geq 1$.

$$(e) \quad \begin{cases} x = 5t \cos 40^\circ \\ y = 5t \sin 40^\circ - 5t^2 + 1.5 \end{cases}$$

$$\Rightarrow \begin{cases} \dot{x} = 5 \cos 40^\circ \\ \dot{y} = 5 \sin 40^\circ - 10t \end{cases}$$

$$\begin{aligned} \bullet \quad \dot{y} &= 0 \\ \Rightarrow 5 \sin 40^\circ - 10t &= 0 \\ 10t &= 5 \sin 40^\circ \\ t &= \frac{\sin 40^\circ}{2} \end{aligned} \quad \left. \vphantom{\begin{aligned} \bullet \quad \dot{y} &= 0 \\ \Rightarrow 5 \sin 40^\circ - 10t &= 0 \\ 10t &= 5 \sin 40^\circ \\ t &= \frac{\sin 40^\circ}{2} \end{aligned}} \right\} \text{--- (1)}$$

$$\star h = y_{t = \frac{\sin 40^\circ}{2}} = 5 \left(\frac{\sin 40^\circ}{2} \right) \sin 40^\circ - 5 \left(\frac{\sin^2 40^\circ}{2} \right) + 1.5 \quad \left. \vphantom{h = y_{t = \frac{\sin 40^\circ}{2}}} \right\} \text{--- (1)}$$

$$\approx 2.0$$

$$\bullet \quad y = 0$$

$$5t \sin 40^\circ - 5t^2 + 1.5 = 0$$

$$5t^2 - 5t \sin 40^\circ - 1.5 = 0$$

$$t = \frac{5 \sin 40^\circ \pm \sqrt{(5 \sin 40^\circ)^2 - 4(5)(-1.5)}}{2(5)}$$

$$t = \frac{5 \sin 40^\circ + \sqrt{25 \sin^2 40^\circ + 30}}{10} \quad \text{--- (1) ---} \quad \text{--- 15 ---}$$

(or decimal roundly)

$$* \quad x = 5 \left(* \right) \times \cos 40^\circ$$

$$\boxed{x \approx 3.7} \quad \text{--- (1) ---}$$

Question 14

$$\begin{aligned} a) (1+kx)^n &= {}^nC_0 + {}^nC_1 kx + {}^nC_2 (kx)^2 + \dots \\ &= 1 + nkx + \frac{n!}{2!(n-2)!} (kx)^2 + \dots \\ &= 1 + nkx + \frac{n(n-1)}{2} (kx)^2 + \dots \quad \text{--- (A)} \end{aligned} \quad \text{--- (1)}$$

Given $(1+kx)^n = 1 + 40x + 120k^2x^2$ --- (B)

$$\begin{aligned} \Rightarrow \text{--- (A)} &= \text{--- (B)} \\ nk &= 40 \quad \& \quad \frac{n(n-1)}{2} = 120 \quad \text{--- (1)} \\ \text{sub.} & \\ k &= \frac{40}{n} \\ \boxed{k = \frac{5}{2}} & \quad \text{--- (2)} \end{aligned}$$

b) (1) Let $x = \cos \theta$

$$\cos 4\theta = 0$$

$$4\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2} \quad \text{--- (1)}$$

$$\theta = \frac{\pi}{8}, \frac{3\pi}{8}, \frac{5\pi}{8}, \frac{7\pi}{8}$$

$$\text{So } x = \cos \frac{\pi}{8}, \cos \frac{3\pi}{8}, \cos \frac{5\pi}{8}, \cos \frac{7\pi}{8} \quad \text{--- (1)}$$

$$(11) \quad \alpha \beta \Delta S = \frac{c}{a}$$

-11-

$$\cos \frac{\pi}{8} \cos \frac{3\pi}{8} \cos \frac{5\pi}{8} \cos \frac{7\pi}{8} = \frac{1}{8} \quad - (1)$$

$$\cos \frac{\pi}{8} \cos \frac{3\pi}{8} \cos \left(\pi - \frac{3\pi}{8} \right) \cos \left(\pi - \frac{\pi}{8} \right) = \frac{1}{8}$$

$$= \cos^2 \left(\frac{\pi}{8} \right) \left(-\cos^2 \left(\frac{3\pi}{8} \right) \right) = \frac{1}{8}$$

$$\left(\cos \left(\frac{\pi}{8} \right) \cos \left(\frac{3\pi}{8} \right) \right)^2 = \frac{1}{8}$$

$$\cos \frac{\pi}{8} \cos \frac{3\pi}{8} = \frac{1}{\sqrt{8}} \text{ or } \frac{1}{2\sqrt{2}} \quad - (1)$$

$$\begin{aligned} e) \quad x &= \tan^2 \theta \\ \frac{dx}{d\theta} &= 2 \sec^2 \theta \tan \theta \\ dx &= 2 \sec^2 \theta \tan \theta \end{aligned} \quad \left. \begin{array}{l} \tan^2 \theta = 0 \\ \theta = 0 \\ \tan \theta = 1 \\ \theta = \frac{\pi}{4} \end{array} \right\} - (1)$$

$$\int_0^{\frac{\pi}{4}} \frac{1 - \tan^2 \theta}{\cancel{\tan \theta} (1 + \tan^2 \theta)^{\cancel{2}}} \times \cancel{2 \sec^2 \theta \tan \theta} d\theta$$

$$= 2 \int_0^{\frac{\pi}{4}} \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} d\theta \quad - (1)$$

$$= 2 \int_0^{\frac{\pi}{4}} \frac{1 - \tan^2 \theta}{\sec^2 \theta} d\theta$$

$$= 2 \int_0^{\frac{\pi}{4}} \left(\frac{1}{\sec^2 \theta} - \frac{\tan^2 \theta}{\sec^2 \theta} \right) d\theta$$

$$\begin{aligned}
&= 2 \int_0^{\frac{\pi}{4}} \left(\cos^2 \theta - \frac{\sin^2 \theta}{\cancel{\cos^2 \theta}} \times \cancel{\cos^2 \theta} \right) \\
&= 2 \int_0^{\frac{\pi}{4}} \cos^2 \theta - \sin^2 \theta \quad - (1) \\
&= 2 \int_0^{\frac{\pi}{4}} \cos 2\theta \\
&= 2 \left[\frac{\sin 2\theta}{2} \right]_0^{\frac{\pi}{4}} \\
&= 2 \left[\frac{1}{2} - 0 \right] \\
&= 1 \quad - (1)
\end{aligned}$$

d) $y^2 = m - 3x$

$$y^2 - m = -3x$$

$$\boxed{x = \frac{m - y^2}{3}} \quad - (1)$$

$$\begin{aligned}
V &= m \int_0^{\sqrt{3}} \left(\frac{m - y^2}{3} \right)^2 dy \quad - (1) \\
&= \frac{m}{9} \int_0^{\sqrt{3}} (m^2 - 2my^2 - y^4) dy \\
&= \frac{m}{9} \left[m^2 y - \frac{2my^3}{3} + \frac{y^5}{5} \right]_0^{\sqrt{3}} \\
&= \frac{m}{9} \left[m^2 \sqrt{3} - \frac{2m^2 \sqrt{3}}{3} + \frac{m^2 \sqrt{3}}{5} - 0 \right] \quad - (1)
\end{aligned}$$

$$= \frac{57}{9} \times \frac{8m^2 m}{15} = \frac{5000}{27} m$$

$$\frac{8m^2 m}{135} = \frac{5000}{27}$$

$$m^{5/2} = 3125$$

$$\boxed{m = 25} - (1)$$